Noor Ul Huda Ajmal

Machine Learning Engineer

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Machine Learning Engineer with 3+ years of experience developing and deploying end-to-end ML systems across computer vision, deep learning, data analytics and applied AI. Experienced in building production ML pipelines, MLOps workflows and real-time inference systems, with additional research experience applying machine learning to Software-Defined Optical Networks and photonic switching systems.

EXPERIENCE

Independent Machine Learning Consultant, Freelance, Remote *July 2022 - Present*

- Design, develop and deploy machine learning solutions for clients across multiple domains.
- Shipped projects spanning computer vision pipelines, deep learning models, MLOps workflows and AI-driven data analytics dashboards.
- Built end-to-end ML systems covering data processing, model training, deployment and monitoring.

Machine Learning Intern, Optical Networking Technologies Lab, NUST, Islamabad *June 2022 - Oct 2022*

- Developed machine learning models for Software Defined Optical Networks (SDON) to predict photonic switch configurations from optical signal wavelengths.
- Performed experiments on network path optimization and noise pattern analysis.
- Contributed to a research project that resulted in a peer-reviewed IEEE publication.

PROJECTS

Machine Learning for Light Path Prediction in Software Defined Optical Networks (SDON)

- Built a multi-label classification model using XGBoost with classifier chains to predict 20 interdependent photonic switch states from 8 optical signal wavelength features, treating it as an inverse control problem to simplify complex optical network management.
- Achieved 95% precision and 96% recall in predicting 20 photonic switch control states, with classifier chains capturing switch dependencies.
- Published the research as a peer-reviewed paper in IEEE Xplore demonstrating how ML-based inverse modeling can simplify complex optical control mechanisms.
- **Tech:** Python, XGBoost, Multi-label Classification, Optical Networks

Machine Learning for Quality of Transmission (QoT) Prediction in Optical Networks

- Explored machine learning approaches for predicting Quality of Transmission (QoT) of optical lightpaths using lightpath and network-level measurements from optical network data.
- Investigated the relationship between optical signal characteristics, network conditions and QoT to understand how data-driven models can be used for lightpath quality assessment.
- Evaluated the potential of ML-based QoT prediction as a data-driven alternative for supporting optical network planning, monitoring and lightpath management.
- **Tech:** Python, Pandas, Scikit-learn, Machine Learning, Optical Networks

AutoVision ModelOps

- Reduced model retraining and deployment overhead by over 70% (estimated manual effort) by building an automated MLOps pipeline that handles everything from data ingestion and augmentation to containerized deployment and version tracking.
- Enabled continuous model improvement with minimal manual intervention by engineering an iterative fine-tuning loop that automatically converts new video data from live deployments into additional training samples, directly supporting accurate object recognition.
- Shortened failure detection and recovery time from hours to minutes by integrating Slack based automated alerts and centralized metadata logging ensuring engineers are notified immediately when training or deployment pipelines encounter issues.
- Forms the core ML pipeline infrastructure supporting the object recognition system of *LivingImage*.
- **Tech:** Python, Azure Custom Vision, Docker, Azure ACI/ACR, GitHub CI/CD, MLOps

Driver Drowsiness Detection using Onboard Camera

- Developed a real-time driver monitoring system using CNN-based drowsiness detection deployed on a Raspberry Pi with an infrared camera.
- Improved detection accuracy by combining facial landmark tracking with a modified PERCLOS algorithm that measures eye closure and yawning patterns achieving the detection accuracy of 88%.
- Integrated embedded sensors with on-device deep learning inference to detect accidents and record events in real time.
- **Tech:** Python, Raspberry Pi, Computer Vision, Deep Learning

Automated Fiber Ring Detection & Measurement

- Built a two-stage hybrid computer vision pipeline combining YOLO-based detection and OpenCV geometric analysis to automate measurement of concentric ring structures in fiber cross-section images, reducing analysis time from hours to minutes per batch despite having only 4-8 unlabeled, distorted microscopy images.
- Handled challenging conditions including uneven lighting, microscope reflections and non-circular ring distortion using HSV segmentation, morphological filtering, contour detection and least-squares ellipse fitting.
- Extracted structural metrics including ring radius, layer thickness and ellipse orientation thereby enabling scalable quality control for fiber manufacturing.
- **Tech:** Python, OpenCV, ONNX, Image Processing, Computer Vision

PUBLICATIONS

IEEE Asia Communications and Photonics Conference (ACP) 2022 *2022*

Multi-labeled Random-Forest Enabled Softwarized Management for Photonics Switching Systems

EDUCATION

Bachelor of Engineering in Software Engineering (BESE), NUST, Islamabad *2019 - 2023*

- GPA: 3.18 / 4.0

SKILLS

Programming: Python, SQL

Machine Learning & AI: Scikit-learn, XGBoost, TensorFlow, Keras, PyTorch, Deep Learning, Model Optimization

Computer Vision: OpenCV, ONNX, Image Processing, Object Detection, Image Classification

MLOps & ML Systems: Docker, MLflow, GitHub CI/CD, REST APIs, Model Deployment

Generative AI: Prompt Engineering, LLM Fine-tuning, Agentic AI

Databases: MySQL, PostgreSQL, Vector Databases (FAISS, ChromaDB, Pinecone)

Data Engineering & Automation: Web Scraping (Selenium, BeautifulSoup), ETL Pipelines, API Integration

Cloud & Systems: Azure, AWS, Linux

Tools: Git